

MEDVEZHYEGORSK, Russian Federation

WMO: 227210

Lat: 62.917N

Lon: 34.433E

Elev: 266

StdP: 14.56

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-21.6	-15.7	-26.5	1.3	-21.5	-20.3	1.8	-15.7	10.0	22.0	9.4	22.2	2.2	290	0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.6	79.8	66.3	76.4	64.5	73.0	62.7	68.6	76.4	66.5	73.6	64.2	70.6	3.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.8	96.2	72.6	63.6	88.8	70.6	61.4	82.1	68.2	33.0	76.6	31.3	73.7	29.6	70.6	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	7.7	7.1	DB	-24.7	84.3	7.8	4.1	-30.3	87.2	-34.9	89.7	-39.3	92.0	-45.0	95.0	(4)
(5)			WB	-24.9	71.7	7.6	3.5	-30.4	74.2	-34.8	76.3	-39.1	78.2	-44.7	80.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.9	13.6	13.5	22.8	34.5	45.6	56.1	62.4	58.1	49.0	38.0	27.5	20.1
	DBStd	19.24	14.42	14.30	9.90	7.34	8.42	7.55	6.39	5.92	6.17	7.24	9.28	12.63
	HDD50	5743	1127	1021	844	466	189	23	2	6	91	374	675	925
	HDD65	10326	1592	1441	1309	915	604	283	125	224	481	837	1125	1390
	CDD50	968	0	0	0	1	52	208	386	259	60	2	0	0
	CDD65	75	0	0	0	0	2	17	45	11	0	0	0	0
	CDH74	567	0	0	0	0	28	154	309	76	0	0	0	0
	CDH80	87	0	0	0	0	3	31	47	6	0	0	0	0
Wind	WSAvg	3.0	3.1	3.1	3.3	2.9	3.1	2.9	2.7	2.8	2.9	3.2	3.1	3.3
Precipitation	PrecAvg	28.2	2.2	1.7	1.6	1.4	2.1	2.6	3.4	3.1	2.7	2.7	2.5	2.3
	PrecMax	35.4	3.7	2.8	3.2	2.6	4.6	5.3	8.5	5.4	7.3	5.4	3.7	4.2
	PrecMin	20.2	0.9	0.4	0.5	0.2	0.7	0.9	1.4	1.1	0.6	1.1	0.3	0.7
	PrecStd	3.9	0.7	0.7	0.6	0.7	0.9	1.1	1.6	1.1	1.3	1.2	0.8	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.0	37.9	48.0	61.2	77.8	84.0	84.6	80.3	68.8	55.7	45.9	41.7
		MCWB	35.3	35.8	38.6	49.2	58.6	66.2	68.2	68.2	59.9	52.9	44.1	40.8
	2%	DB	34.4	35.6	41.3	53.7	70.2	78.2	80.6	76.0	64.4	52.4	43.5	37.7
		MCWB	33.4	34.0	35.0	44.4	55.3	63.6	67.9	64.6	57.4	50.5	42.4	36.6
	5%	DB	33.1	33.7	37.5	49.7	65.1	73.8	77.5	72.4	61.0	50.3	41.3	35.7
		MCWB	32.4	32.7	33.5	41.8	52.7	61.4	66.0	62.9	55.9	48.5	40.6	34.7
	10%	DB	31.7	31.9	35.1	45.6	60.0	69.6	74.2	69.1	58.4	48.2	38.8	34.1
		MCWB	31.2	31.2	32.4	39.3	50.4	59.4	64.4	61.0	54.7	46.6	38.0	33.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.8	36.4	39.6	49.9	61.7	69.5	72.5	70.3	61.6	53.7	45.0	41.1
		MCDB	36.7	37.6	46.2	58.6	73.2	79.0	80.7	76.8	66.5	55.0	45.7	42.0
	2%	WB	33.7	34.3	36.0	45.5	57.5	66.4	69.8	66.7	58.8	51.0	42.5	37.0
		MCDB	34.2	35.1	39.5	52.1	67.8	75.5	77.7	73.7	62.5	52.1	43.2	37.6
	5%	WB	32.5	32.8	34.3	42.5	54.2	63.2	67.6	64.0	56.9	48.8	40.5	35.0
		MCDB	33.0	33.6	36.6	48.4	62.7	71.2	74.5	69.8	59.7	49.9	41.3	35.5
	10%	WB	31.1	31.2	33.0	39.8	51.2	60.1	65.6	61.9	55.2	46.7	38.1	33.4
		MCDB	31.7	32.0	34.8	44.6	59.0	68.0	72.0	67.5	57.9	48.0	38.8	34.0
Mean Daily Temperature Range	5% DB	MDBR	10.4	12.4	14.5	15.4	17.0	16.7	15.6	15.0	12.3	8.0	7.3	8.7
		MCDBR	7.1	8.4	14.1	21.8	25.7	22.5	20.9	20.9	16.7	9.5	7.2	6.7
		MCWBR	7.4	7.9	9.9	13.3	12.7	10.6	9.8	10.7	10.5	7.7	7.1	6.7
	5% WB	MCDBR	6.9	7.9	11.6	19.5	21.3	19.3	18.1	17.2	13.5	8.6	7.0	6.4
		MCWBR	7.0	7.6	8.7	12.8	12.1	10.5	10.0	10.0	9.4	7.6	7.1	6.5
Clear-Sky Solar Irradiance	taub	0.235	0.288	0.314	0.352	0.353	0.346	0.379	0.361	0.333	0.312	0.246	0.210	
	taud	1.993	2.082	2.145	2.278	2.445	2.495	2.420	2.446	2.511	2.390	2.079	1.956	
	Ebn at Noon	122	200	245	263	274	278	264	259	245	199	117	67	
	Edh at Noon	10	22	30	33	31	30	32	28	22	16	10	5	
All-Sky Solar Radiation	RadAvg	61	244	701	1170	1547	1719	1600	1189	671	251	65	26	
	RadStd	8	54	99	118	143	123	183	112	74	36	12	3	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	+1.10	N/A	N/A	N/A	N/A	N/A	N/A	-367	N/A	N/A

Nomenclature: See separate page